

Quantum Mekaniği I Bölüm 4

Tekrar

$$i\hbar \frac{\partial \Psi}{\partial t} = -\hbar^2 \frac{\partial^2 \Psi}{\partial x^2}$$

$$\Psi(x) = U(x)T(t)$$

$$\Rightarrow i\hbar \frac{\partial \Psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \Psi(x) = E$$

$$\Rightarrow i\hbar \frac{\partial U(x)T(t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2 U(x)T(t)}{\partial x^2} + V(x)U(x)T(t) \right] = E U(x)T(t)$$

$$\Rightarrow i\hbar \frac{1}{T(t)} \frac{\partial T(t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 U(x)}{\partial x^2} + V(x) = E$$

$$\Rightarrow i\hbar \frac{\partial T}{T(t)} = E dt \Rightarrow \frac{\partial T}{T(t)} = \frac{1}{i\hbar} E dt \Rightarrow \ln T = \frac{Et}{i\hbar} + \ln C \Rightarrow \left(\frac{\ln T}{\ln C} \right) = \left(\frac{Et}{i\hbar} \right) T = C e^{\frac{Et}{i\hbar}}$$

$$T = C e^{-iEt/\hbar}$$

$$\Psi(x, t) = C e^{-iEt/\hbar} U(x)$$

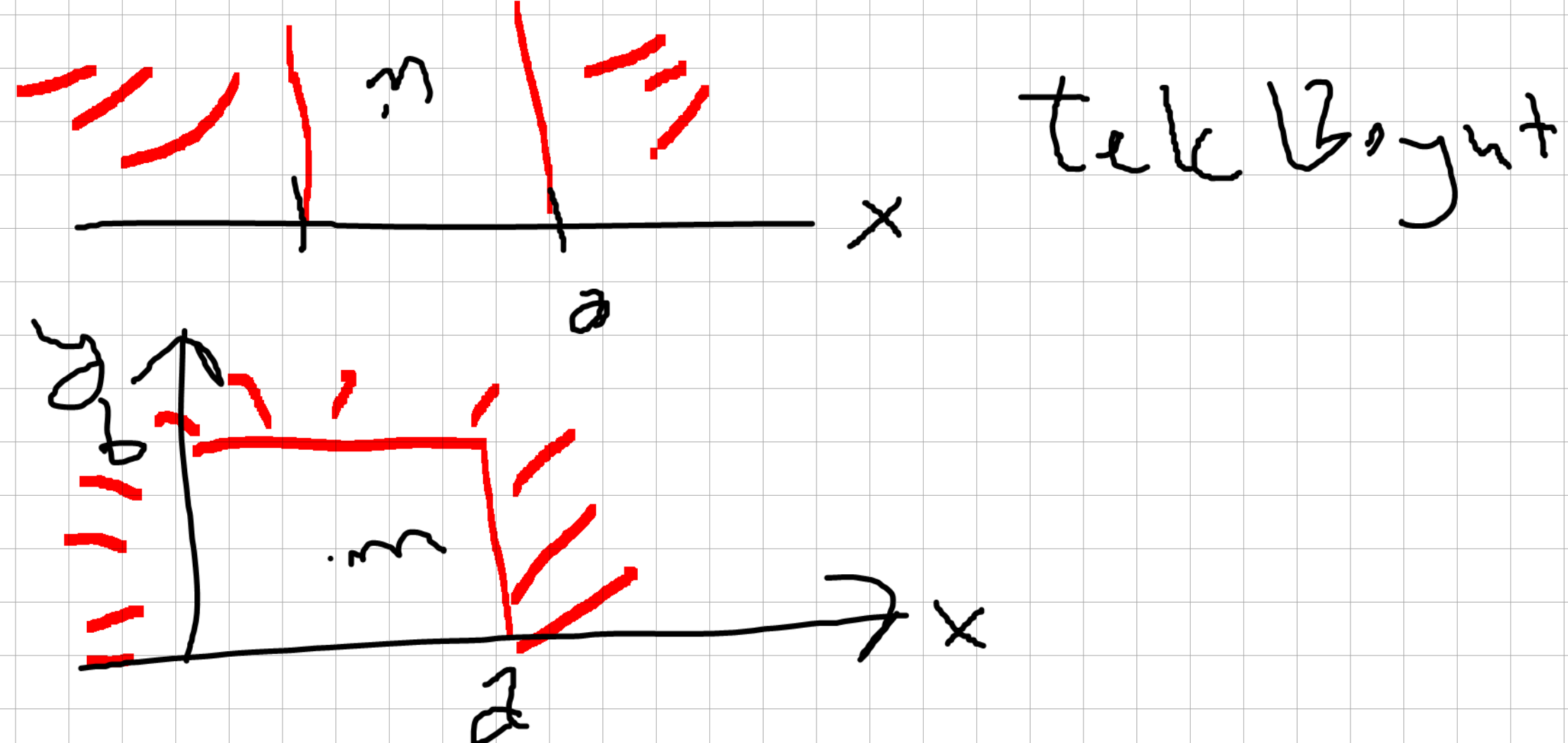
Operatörler $L[f_1(x) + f_2(x)] = Lf_1(x) + Lf_2(x)$ $f_1 + f_2$ lineerdir

$c \in \mathbb{C}$
 $L[c f(x)] = c Lf(x)$ $c \rightarrow \mathbb{C}^*$ karşın antylineer

Bir kutu içerisinde parçacık özbeğen problemi (1 Boyutlu)

$$V(x) \begin{matrix} x \geq a & \infty \\ 0 < x < a & 0 \\ x \leq 0 & \infty \end{matrix}$$

$$V(x, y) \begin{matrix} x > a & y > b & \infty \\ x < 0 & y < 0 & \infty \\ 0 < x < a & 0 < y < b & 0 \end{matrix}$$



tek boyut

$$-\frac{\hbar^2}{2m} \frac{\partial^2 U(x)}{\partial x^2} + V(x) = E U(x)$$

$$V(x) = 0 \quad -\frac{\hbar^2}{2m} \frac{\partial^2 U}{\partial x^2} = E U \Rightarrow \frac{\partial^2 U}{\partial x^2} = -\frac{2mE}{\hbar^2} U \Rightarrow \frac{\partial^2 U}{\partial x^2} + \frac{2mE}{\hbar^2} U = 0$$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

$$D^2 + k^2 = 0 \quad D = \sin, \cos, e^{i\cdot}, e^{-i\cdot}$$

$$\Rightarrow D = A \cos kx + B \sin kx$$

$$D' = -Ak \sin kx + Bk \cos kx$$

$$D'' = -Ak^2 \cos kx - Bk^2 \sin kx$$

$$x \in [0, a]$$

$$x=0 \Rightarrow 0 \\ x=a \Rightarrow 0$$

$$x=0 \Rightarrow D'' = -Ak^2 + 0 \Rightarrow A=0$$

$$D = B \sin kx$$

$$x=a \Rightarrow -Bk^2 \sin ka = 0$$

$$D = B \sin \frac{n\pi x}{a}$$

$$0 \quad n\pi \quad 2n\pi \quad 3n\pi \quad \dots$$

$$\Rightarrow n\pi = ka \Rightarrow$$

$$k = \frac{n\pi}{a} \quad k^2 = \frac{2mE}{\hbar^2}$$

$$\Rightarrow \frac{n^2 \pi^2}{a^2} = \frac{2mE}{\hbar^2} \Rightarrow E = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

$$B=? \quad B \sin ka \Rightarrow B \sin n \frac{12x}{2}$$

$$\int dx U_n^* U_n = 1 \Rightarrow$$

$$\int dx B^2 \sin^2 \frac{n12x}{2} = 1$$

$$\Rightarrow B^2 \int_{-\frac{a}{2}}^{\frac{a}{2}} dx \sin^2 \frac{n12x}{2} = 1$$

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$

$$y \rightarrow x \Rightarrow \cos(2x) = \cos^2 x - \sin^2 x$$

$$\Rightarrow \sin^2 x = \cos^2 x - \cos 2x$$

$$1 - \sin^2 x \Rightarrow 2 \sin^2 x = 1 - \cos 2x \Rightarrow \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\frac{B^2}{2} \int_0^a dx (1 - \cos 2x)$$

$$\frac{B^2}{2} \int_0^a dx (1 - \cos 2 \frac{n12x}{2})$$

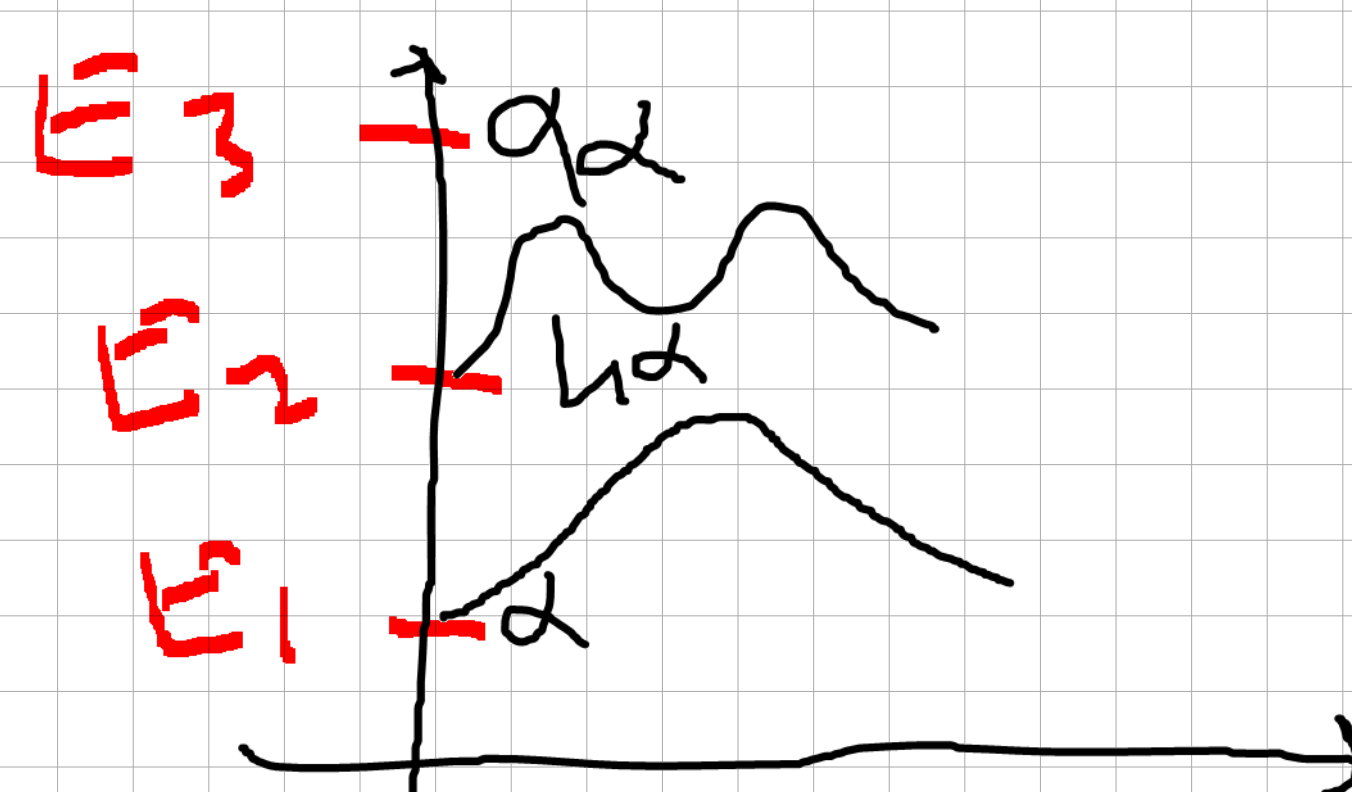
$$\frac{B^2}{2} a = 1 \Rightarrow B^2 = \frac{2}{a} \Rightarrow B = \sqrt{\frac{2}{a}}$$

$$\boxed{B \sin kx} \quad k \vee \quad B \vee \quad E \vee$$

$$E = \frac{n^2 12^2 \hbar^2}{2ma^2}$$

$$k = \frac{n12}{a} \quad k^2 = \frac{2mE}{\hbar^2}$$

$$E_1 = \frac{12^2 \hbar^2}{2ma^2} \quad E_2 = 4E_1 \quad E_3 = 9E_1$$



$$\psi(x) = \sum_n A_n U_n(x) = A_{\text{norm}} \text{ Post}$$

$$\psi^* \psi = A_n^* A_n = |A_n|^2$$

